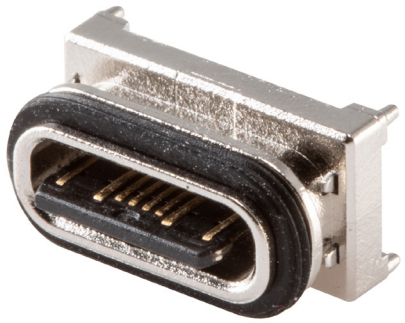
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# MODEL: UJ20-C-V-SMT-TR-68 | DESCRIPTION: USB RECEPTACLE

## FEATURES

- USB Type C receptacle
- IP68
- USB 2.0
- o-ring accessory
- tape & reel packaging



## SPECIFICATIONS

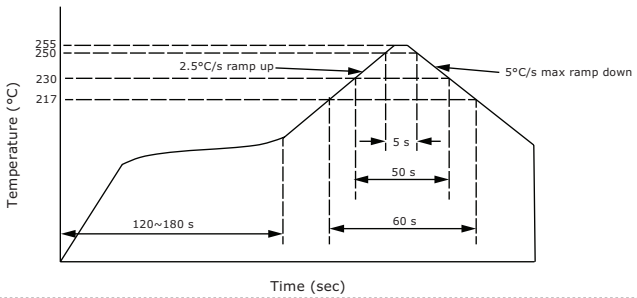
| parameter | conditions/description | min | typ | max | units |
|---|---|---|---|---|---|
| USB standard | USB 2.0 | | | | |
| rated input voltage | | | | 20 | Vdc |
| rated input current | as per EIA-364-70, Method B<br>power pins collectively (A1, A4, A9, A12, B1, B4, B9, B12)<br>VCONN pins individually (A5, B5)<br>all other signal pins individually | | | <br>5.0<br>1.25<br>0.25 | <br>A<br>A<br>A |
| contact resistance[1] | between terminals and mating plug | | | 40 | mΩ |
| insulation resistance | between adjacent contacts | 100 | | | MΩ |
| voltage withstand | between adjacent contacts | 100 | | | Vac |
| insertion force | | 5 | | 20 | N |
| withdrawal force | | 8 | | 20 | N |
| life | | | 10,000 | | cycles |
| operating temperature | | -40 | | 85 | °C |
| flammability rating | UL94V-0 | | | | |
| RoHS | yes | | | | |
| IP level | IP68 | | | | |

Note: 1. When measured at 20 mV / 100 mA.
2. All specifications measured at 10~35°C, humidity at 45~85%, under standard atmospheric pressure, unless otherwise noted.

## SOLDERABILITY

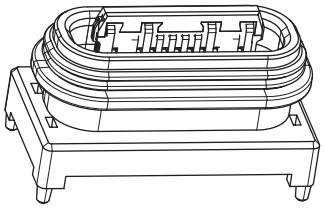
| parameter | conditions/description | min | typ | max | units |
|---|---|---|---|---|---|
| reel storage | at relative humidity 65~70% | 25 | | 30 | °C |
| reflow soldering[3] | see reflow profile | | 255 | 260 | °C |

Note: 3. It is recommended to reflow solder within 3 months from opening vacuum packaging at a temperature of 25~30°C & relative humidity 65~70%.

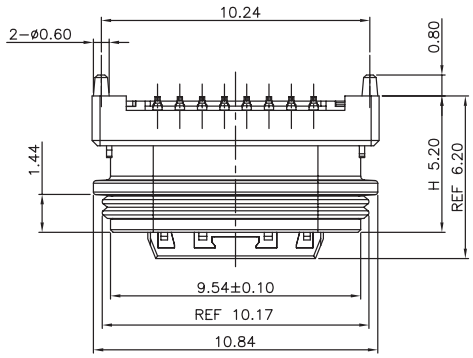


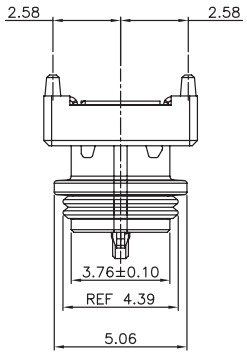
## MECHANICAL DRAWING

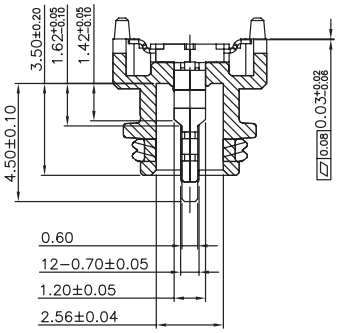
units: mm
tolerance:
X ±0.50 mm
X.X ±0.30 mm
X.XX ±0.20 mm
X.XXX ±0.100 mm
unless otherwise specified



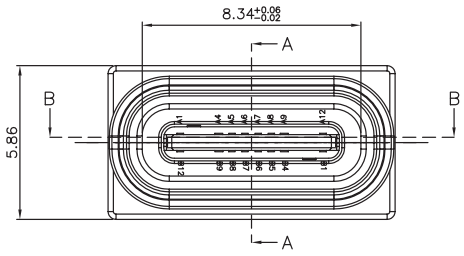
| DESCRIPTION | MATERIAL | PLATING/COLOR |
|---|---|---|
| insulator | LCP (UL94V-0) | black |
| contact | copper alloy<br>t=0.15 mm | contact area: gold flash over tin-nickel<br>solder area: tin over nickel |
| shell | stainless steel<br>t=0.25 mm | nickel |
| o-ring (OR-120) | silicone | black |

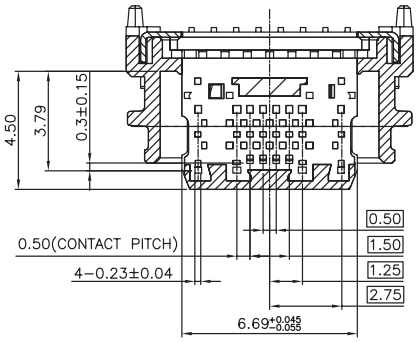


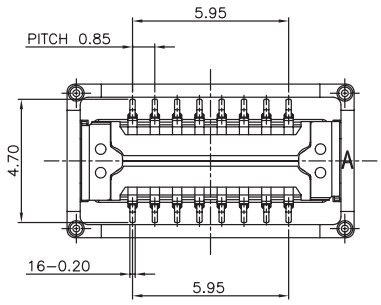




SECTION A−A

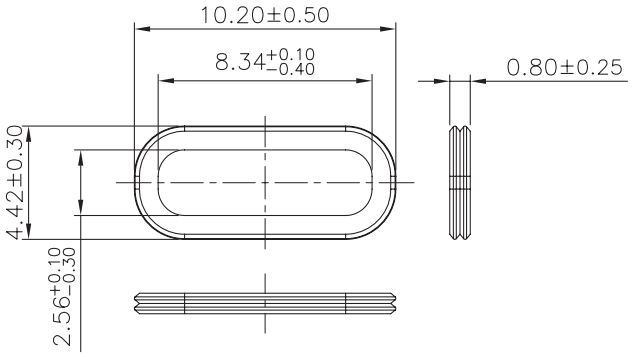



SECTION B−B




### O-ring (P/N: OR-120)

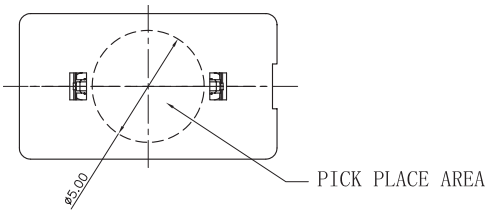


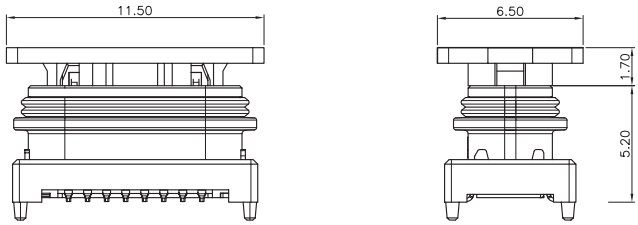
## MECHANICAL DRAWING (CONTINUED)

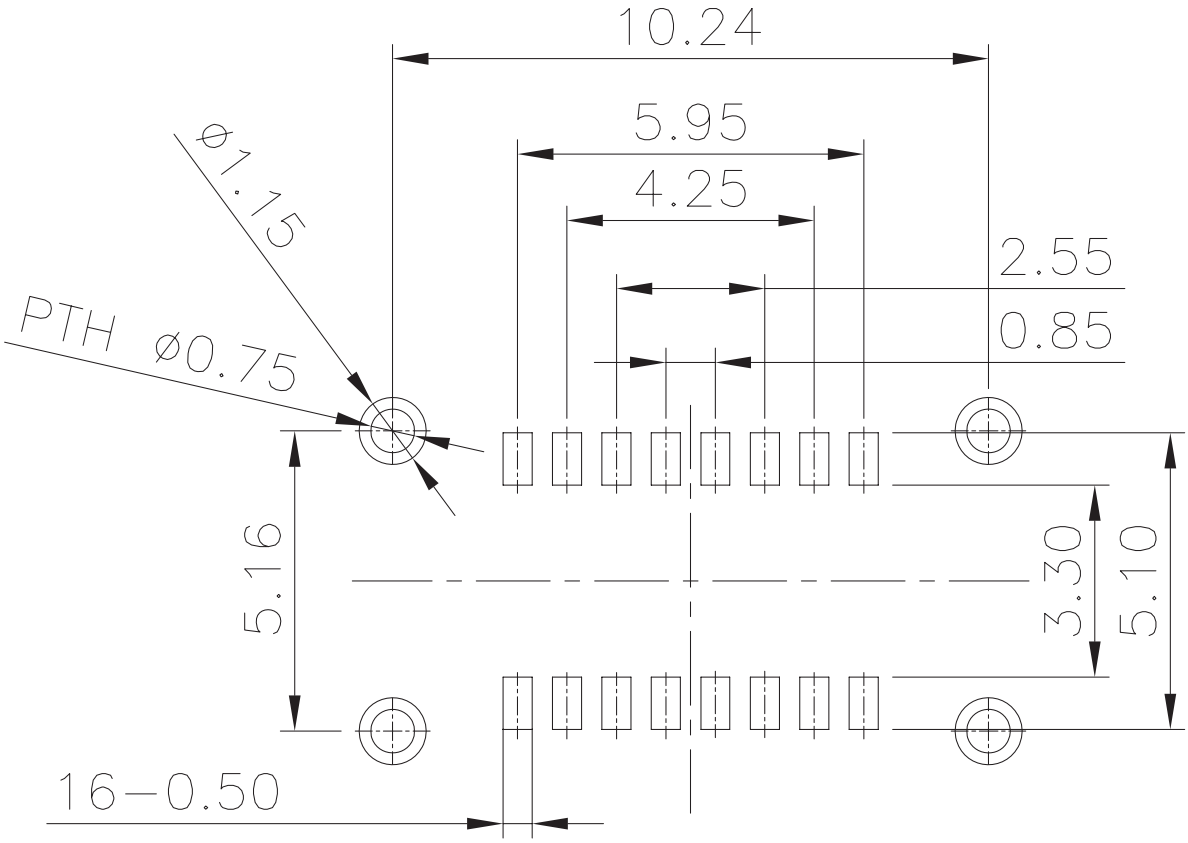
units: mm
PCB tolerance: ±0.05 mm

| PIN CONNECTIONS | | | | | |
|---|---|---|---|---|---|
| PIN | SIGNAL NAME | MATING SEQUENCE | PIN | SIGNAL NAME | MATING SEQUENCE |
| A1 | GND | FIRST | B12 | GND | FIRST |
| A4 | $V_{BUS}$ | FIRST | B9 | $V_{BUS}$ | FIRST |
| A5 | CC1 | SECOND | B8 | SBU2 | SECOND |
| A6 | Dp1 | SECOND | B7 | Dn2 | SECOND |
| A7 | Dn1 | SECOND | B6 | Dp2 | SECOND |
| A8 | SBU1 | SECOND | B5 | CC2 | SECOND |
| A9 | $V_{BUS}$ | FIRST | B4 | $V_{BUS}$ | FIRST |
| A12 | GND | FIRST | B1 | GND | FIRST |
| SHELL | | GND | SHELL | | GND |

**with cap**

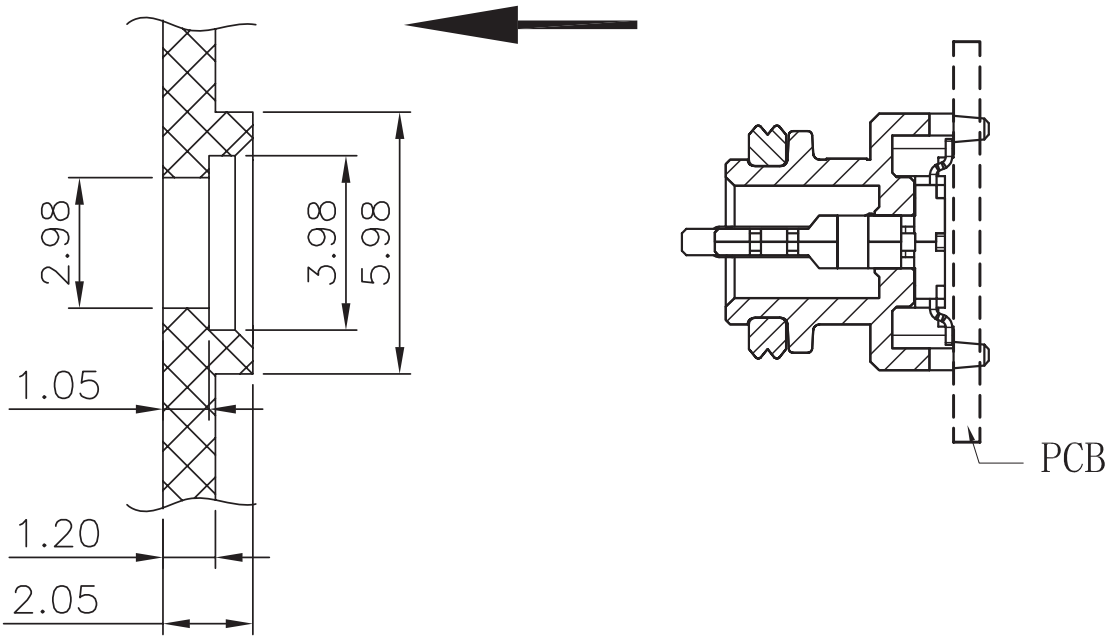


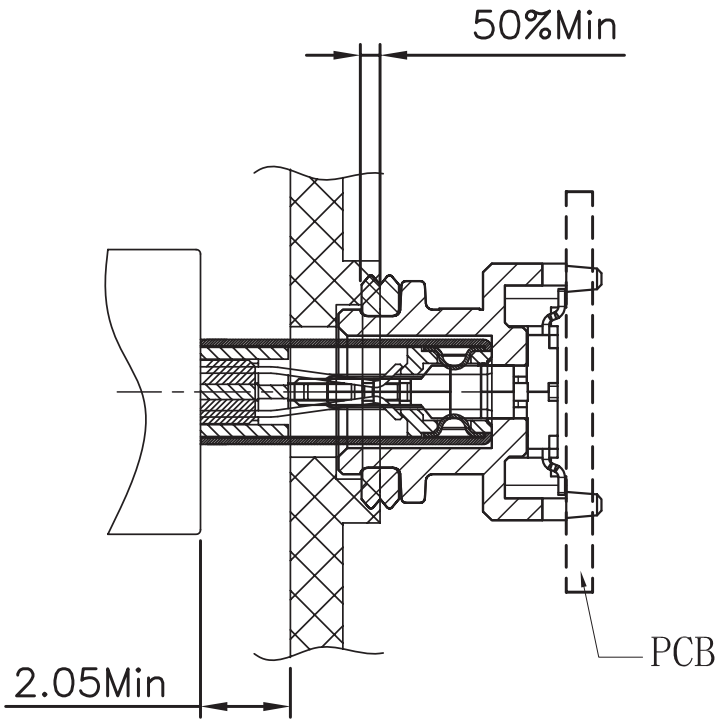




Recommended PCB Layout
Top View

## ASSEMBLY REFERENCE

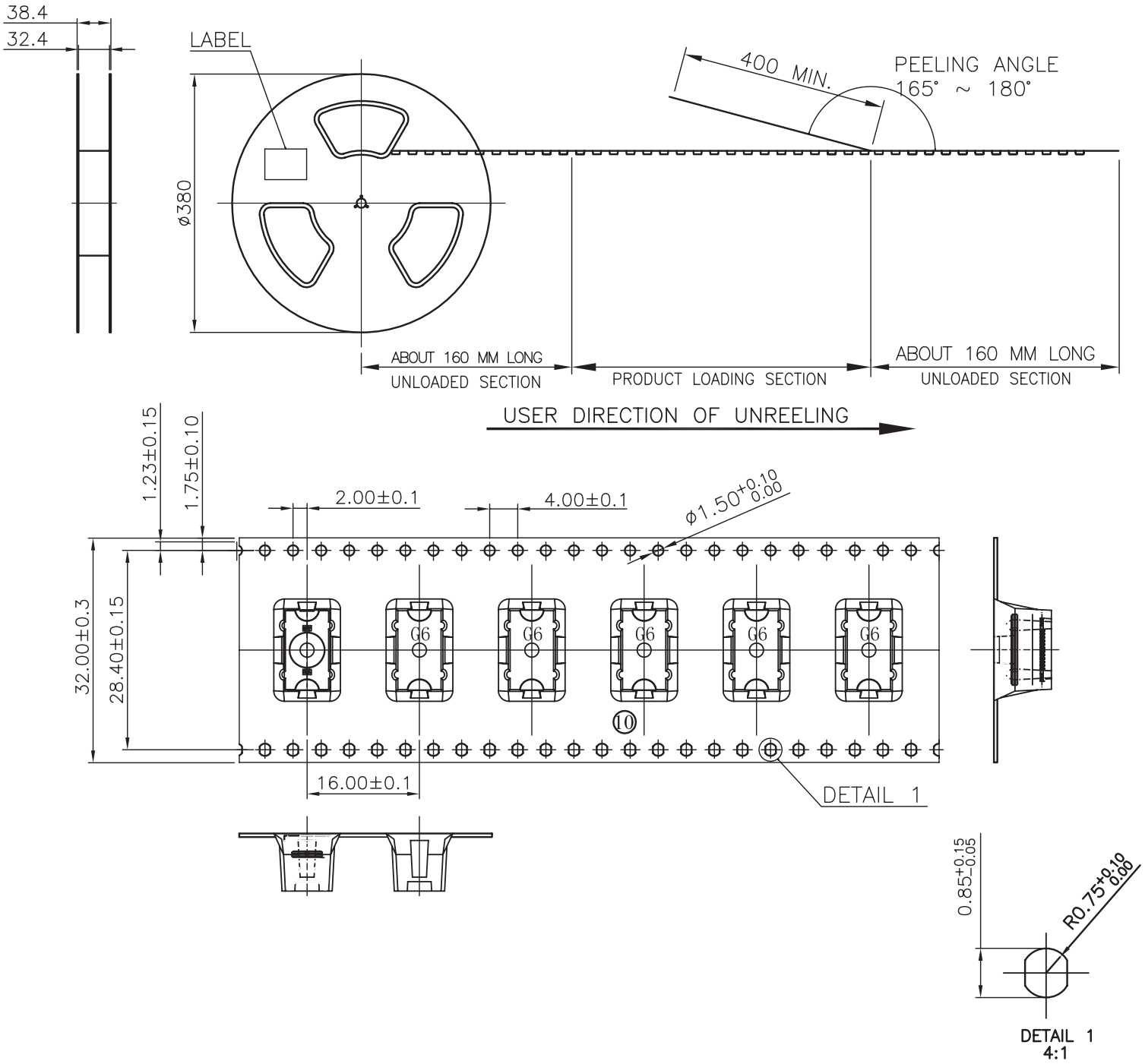


## PACKAGING

units: mm

Reel Size: Ø380 mm
Reel QTY: 650 pcs per reel
Carton Size: 405 x 405 x 330 mm
Carton QTY: 5,200 pcs



## REVISION HISTORY

| rev. | description | date |
|---|---|---|
| 1.0 | initial release | 08/23/2023 |
| 1.01 | CUI Devices rebranded to Same Sky | 09/12/2024 |
| 1.02 | added o-ring part number | 10/08/2025 |

The revision history provided is for informational purposes only and is believed to be accurate.

same sky

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